

Chapter 24

Lecture 24: Stone–Čech compactification. Complete metric spaces.

Theorem 24.0.1. [Recall: Stone–Čech compactification] Let X be a completely regular space. Then there exists a compactification Y of X such that every bounded continuous function $f : X \rightarrow \mathbb{R}$ extends uniquely to a continuous function $\bar{f} : Y \rightarrow \mathbb{R}$. (Y is called the Stone–Čech compactification of X)

Lemma 24.0.1. Let $A \subseteq X$. Let $f : A \rightarrow Z$ be continuous and Z be Hausdorff. Then there exists at most one extension of f to a continuous function $g : \bar{A} \rightarrow Z, g|_A = f$.

Proof

To be done.

Remark The uniqueness part of Theorem 24.0.1 follows from this lemma.

Theorem 24.0.2. Let X be completely regular and Y be a compactification of X such that Y satisfies the Stone–Čech property 24.0.1. Then any continuous map $f : X \rightarrow C$ where C is a compact Hausdorff space can be extended uniquely to a continuous map $g : Y \rightarrow C$.

Proof

To be done.

Theorem 24.0.3. Let Y_1, Y_2 be compactifications of a completely regular space X . Suppose both Y_1 and Y_2 satisfy the Stone–Čech property. Then there exist homeomorphisms $h : Y_1 \rightarrow Y_2$.

Proof

To be done.

24.1 Complete Metric Spaces

Definition 24.1.1. Let (X, d) be a metric space. A sequence $\{x_n\} \subseteq X$ is called a Cauchy sequence if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_n, x_m) < \varepsilon$.

Property 24.1.1. Every convergent sequence is a Cauchy sequence.

Proof To be done.

Definition 24.1.2. A metric space (X, d) is called complete if every Cauchy sequence in X converges to a limit in X .

Example 24.1.1 (non-example). $(0, 1)$ with the usual metric. Then the sequence $x_n = \frac{1}{n}$ is a Cauchy sequence that does not converge in $(0, 1)$.

Example 24.1.2 (non-example). $\mathbb{Q}$ with the usual metric is not complete. For example, the sequence defined by the decimal approximations of $\sqrt{2}$ is a Cauchy sequence in $\mathbb{Q}$ that does not converge to a rational number.

Property 24.1.2. If $A \subseteq X$ is closed and (X, d) is complete, then A is complete.

Property 24.1.3. Let (X, d) be a metric space. Take $\bar{d} = \min\{d, 1\}$. Then (X, d) is complete implies $(X, \bar{d})$ is complete, and conversely.

Lemma 24.1.1. *A metric space X is complete if and only if every Cauchy sequence has a convergent subsequence.*

Proof

“ $\Rightarrow$ ”: Trivial.

“ $\Leftarrow$ ”: To be done.

Theorem 24.1.1. $\mathbb{R}^k$ is complete in d_1, d_2, d_∞ .

Proof

We show this for d_∞ .

Let $\{x_n\} \subseteq \mathbb{R}^k$ be a Cauchy sequence in d_∞ . Then $\{x_n\}$ is bounded, i.e., there exists $M > 0$ such that for all n , $d_\infty(x_n, 0) < M$. So such sequence lies in a compact set $[-M, M]^k$. We know in metric spaces, compactness is equivalent to sequential compactness. Thus there exists a convergent subsequence $\{x_{n_k}\}$ that converges to some $x \in [-M, M]^k$. By the lemma, $\{x_n\}$ converges to x .

All these metrics are equivalent, so $\mathbb{R}^k$ is complete in d_1, d_2 as well. ■

Lemma 24.1.2. *Let $X = \prod_{\alpha \in J} X_\alpha$. Let (x_n) be a sequence in X . Then (x_n) converges to $x \in X$ if and only if for all $\alpha \in J$, the sequence of α -th coordinates $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$ in X_α .*

Proof

“ $\Rightarrow$ ”: π_α is continuous for all $\alpha \in J$. Then it preserves convergence.

“ $\Leftarrow$ ”: Suppose $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$ for all $\alpha \in J$. Let U be an open neighborhood of x . Then there exists a finite set $K \subseteq J$ and open sets $U_\alpha \subseteq X_\alpha$ for all $\alpha \in K$ such that $x \in \prod_{\alpha \in K} U_\alpha \times \prod_{\alpha \in J \setminus K} X_\alpha \subseteq U$. Since $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$, for each $\alpha \in K$, there exists $N_\alpha \in \mathbb{N}$ such that for all $n \geq N_\alpha$, $\pi_\alpha(x_n) \in U_\alpha$. Let $N = \max_{\alpha \in K} N_\alpha$. Then for all $n \geq N$, $x_n \in U$. Thus (x_n) converges to x .

Theorem 24.1.2. *There is a metric on $\mathbb{R}^\omega$ that makes it complete.*

Proof

Define $\bar{d}(a, b) = \min\{1, |a - b|\}$ for all $a, b \in \mathbb{R}$. Define $D(x, y) = \sup_{i \in \mathbb{N}} \{\frac{\bar{d}(x_i, y_i)}{i}\}$ for all $x = (x_i), y = (y_i) \in \mathbb{R}^\omega$. Then D is a metric on $\mathbb{R}^\omega$ that induces the product topology (I think it is proved in some lectures ago, if it is not proved, it is an exercise).

Let $\{x_n\} \subseteq \mathbb{R}^\omega$ be a Cauchy sequence in D . Because

$$\bar{d}(\pi_i(x), \pi_i(y)) \leq i \cdot D(x, y) \tag{24.1}$$

we see that for fixed i , $\{\pi_i(x_n)\}$ is a Cauchy sequence in $\mathbb{R}$, so it converges, say to a_i ($\bar{d}$ is equivalent to $|\cdot|$ in the sense of convergence/completeness as the property says). Then the sequence x_n converges to $a = (a_i) \in \mathbb{R}^\omega$. ■

24.2 Uniform Metric

Let (Y, d) be a metric space. Let $\bar{d} = \min\{d, 1\}$. We take the set of functions Y^J . There is a metric $\bar{\rho}$ on Y^J defined by $\bar{\rho}(x, y) = \sup_{\alpha \in J} \{\bar{d}(x_\alpha, y_\alpha)\}$. This is a metric (exercise). It is called the uniform metric.

Example 24.2.1. *Let $J = [0, 1]$ and $Y = \mathbb{R}$. Define $\bar{d}(f, g) = \sup_{x \in [0, 1]} \{\min(1, |f(x) - g(x)|)\}$.*

Theorem 24.2.1. *If Y is complete with respect to d , then Y^J is complete in the uniform metric $\bar{\rho}$ corresponding to d .*

Theorem 24.2.2. *Let X be a topological space and let (Y, d) be a metric space. The set $C(X, Y)$ of continuous functions from X to Y is closed in Y^X under the uniform metric. So is the set $B(X, Y)$ of bounded functions. Therefore, if Y is complete, these spaces are complete in the uniform metric.*